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COMPUTER APPLICATION

Radiomics based on brain-to-tumor interface enables prediction of metastatic tumor type of brain metastasis: a proof-of-concept study

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Abstract
Background Early and accurate identification of the metastatic tumor types of brain metastasis (BM) is essential for appropriate treatment and management.
Methods A total of 450 patients were enrolled from two centers as a primary cohort who carry 764 BMs originated from non-small cell lung cancer (NSCLC, patient = 173, lesion = 187), small cell lung cancer (SCLC, patient = 84, lesion = 196), breast cancer (BC, patient = 119, lesion = 200), and gastrointestinal cancer (GIC, patient = 74, lesion = 181). A third center enrolled 28 patients who carry 67 BMs (NSCLC = 24, SCLC = 22, BC = 10, and GIC = 11) to form an external test cohort. All patients received contrast-enhanced T1-weighted (T1CE) and T2-weighted (T2W) MRI scans at 3.0 T before treatment. Radiomics features were calculated from BM and brain-to-tumor interface (BTI) region in the MRI image and screened using least absolute shrinkage and selection operator (LASSO) to construct the radiomics signature (RS). Volume of peritumor edema (VPE) was calculated and combined with RS to create a joint model. Performance of the models was assessed by receiver operating characteristic (ROC).
Results The BTI-based RS showed better performance compared to BM-based RS. The combined models integrating BTI features and VPE can improve identification performance in AUCs in the training (LC/NLC vs. SCLC/NSCLC vs. BC/GIC, 0.803 vs. 0.949 vs. 0.918), internal validation (LC/NLC vs. SCLC/NSCLC vs. BC/GIC, 0.717 vs. 0.854 vs. 0.840), and external test (LC/NLC vs. SCLC/NSCLC vs. BC/GIC, 0.744 vs. 0.839 vs. 0.800) cohorts.
Conclusion This study indicated that BTI-based radiomics features and VPE are associated with the metastatic tumor types of BM.

Keywords Brain metastasis · BTI · Metastatic tumor type · Radiomics

Abbreviations

BM	Brain metastasis
BTI	Brain-to-tumor interface
VPE	Volume of peritumoral edema

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T1CE	Contrast-enhanced T1 weighted
T2W	T2 weighted
RS	Radiomics signature
ROC	Receiver operating characteristic
AUC	Area under the ROC curve
BC	Breast cancer
CI	Confidence interval
ACC	Accuracy
SPE	Specificity
SEN	Sensitivity
GIC	Gastrointestinal cancer
SCLC	Small cell lung cancer
NSCLC	Non-SCLC
ROI	Region of interest

Introduction

The metastasis of malignant tumors to brain has been a devastating clinical reality and strongly associated with poor survival outcomes despite advances in systemic therapies for patients with brain metastasis (BM), due to the unique microenvironment of brain [1]. Approximately 20–50% of cancer patients will develop BM depending on the types of primary tumor [2]. The majority of BM usually occurs in patients with lung, breast, gastrointestinal, and melanoma cancer, with the incident rate of 39–56%, 13–30%, 6–9%, and 6–11%, respectively [3]. In current clinical practice, identification of the metastatic tumor types of BM remains challenging. Routinely used clinical radiology (e.g., CT and MRI) can provide insufficient diagnostic power to accurately identify the origin of BM, and thus, extensive steps (including PET imaging and biopsy) are required [4]. However, even after comprehensive diagnosis, still approximately 2–14% patients are carrying BM with unknown primary site [5, 6]. Patients with such conditions typically have a poor prognosis, and the treatment approach is primarily palliative [7]. Therefore, there is an urgent need for a quantitative method to accurately assist in identifying the origin of BM to optimize clinical decision-making [8].

In recent years, radiomics has attracted much attention for the ability to provide quantitative analysis on medical imaging and preoperative assessment of tumor microenvironment [9]. Many studies showed that the radiomics enables the extraction of a large mass of quantitative features from BM and reflection meaningful multifaceted biologic characteristics of BM [10]. Two investigations explored the feasibility of radiomics for prediction of the metastatic tumor types of BM [11, 12]. Another recent effort developed a machine learning model to classify BMs with primary lung and non-lung origin [13]. However, the studies are hindered since they just simply extracted features and build models based on the whole tumor region of BM, which ignored the unique brain microenvironment.

The interface between brain parenchyma and tumor, known as brain-to-tumor interface (BTI), has been demonstrated to be the region where brain tumors interact with endocranial brain cells and immune system [14]. Previous researches have revealed that MRI-based radiomics features derived from BTI are strongly associated with brain invasion status [15, 16], tumor grading [17], and the distinguishment of brain abscess from necrotic glioblastoma [18]. For BM, the BTI-based radiomics was proved to be associated with gene mutation status and therapeutic response [19, 20], while whether the BTI is correlated with the metastatic tumor types of BM is yet to be investigated, and such study may provide new insights and widen the understanding of the microenvironment of BM. Our study provides a non-invasive tool based on the BTI region of BM to assist in identifying the primary tumor site more efficiently for all BM patients, especially for those whose primary tumor location remains unknown even after comprehensive diagnostic evaluation.

Materials and methods

Patients

The retrospective study was approved by the ethics committee of our hospital (2024K022). The informed consent form was waived due to the retrospective nature. A total of 478 patients from three clinical centers who underwent T1-weighted contrast-enhanced (T1CE) and T2-weighted (T2W) brain MRI scans were enrolled between January 2017 and December 2023 for radiomics analyses. The inclusion criteria were: (i) patients underwent T1CE and T2W MRI scans before treatment and (ii) location of the primary tumor was confirmed by medical history and imaging examination, and a biopsy was performed to confirm the histopathology type. Exclusion criteria were: (i) relevant clinical indicators are not documented; (ii) poor image quality of brain MRI; and (iii) patient received specific treatment for BM prior to image acquisition (radiation, surgery, chemotherapy, or targeted therapy). Clinical characteristics for the patients were selected from the patient's medical record.

In the primary cohort (450 patients from centers 1 and 2), a total of 764 BM lesions were included, 187 from NSCLC, 196 from SCLC, 200 from BC, and 181 from GIC. The lesions were randomly divided into a training cohort (507 lesions, NSCLC = 124, SCLC = 130, BC = 133, and GIC = 120) and an internal validation cohort (257 lesions, NSCLC = 63, SCLC = 66, BC = 67, and GIC = 61) by stratified sampling at a 2:1 ratio. In the external test cohort (28 patients from center 3), a total of 67 lesions (NSCLC = 24, SCLC = 22, BC = 10, and GIC = 11) were used to independently validate our findings. Figure 1 shows that a total of 478 patients were eventually enrolled from three centers.

MRI acquisition

Patients from center 1 were scanned with a 3.0-T MRI scanner (Philips Ingenia, Netherlands). Patients from center 2 were scanned with a 3.0-T MRI scanner (Siemens Verio, Erlangen). Patients from center 3 were scanned with a 3.0-T MRI scanner (General Electric Discovery, USA). The T1CE MR images were obtained after injection of gadolinium-diethylenetriamine penta-acetic acid (Gd-DTPA) at the injection speed of 3 mL/s and the dose of 0.2 mL/kg. Detailed scanning parameters of the MRI sequences are shown in Appendix S1.

Tumor segmentation and brain-to-tumor interface generation

For each MRI sequence, a radiologist with 10 years of working experience drew the regions of interest (ROIs) slice by slice along the boundary of BM and the peritumoral edema area (PEA) in T1CE and T2W MRI images and it is stored in a NII format. The ROI segmentation was performed using the open-source ITK-Snap software (version 3.6.0; www.itksnap.org). The segmentation process was under the supervision of a senior radiologist with 15 years of experience to make sure the segmentations were correct. The two readers were blinded to the pathologic results of the patients. Python script (version 3.8.3) was used to generate mask ROIs of the BTI region. The required number of pixels for expansion is first determined by converting the physical distance (mm)

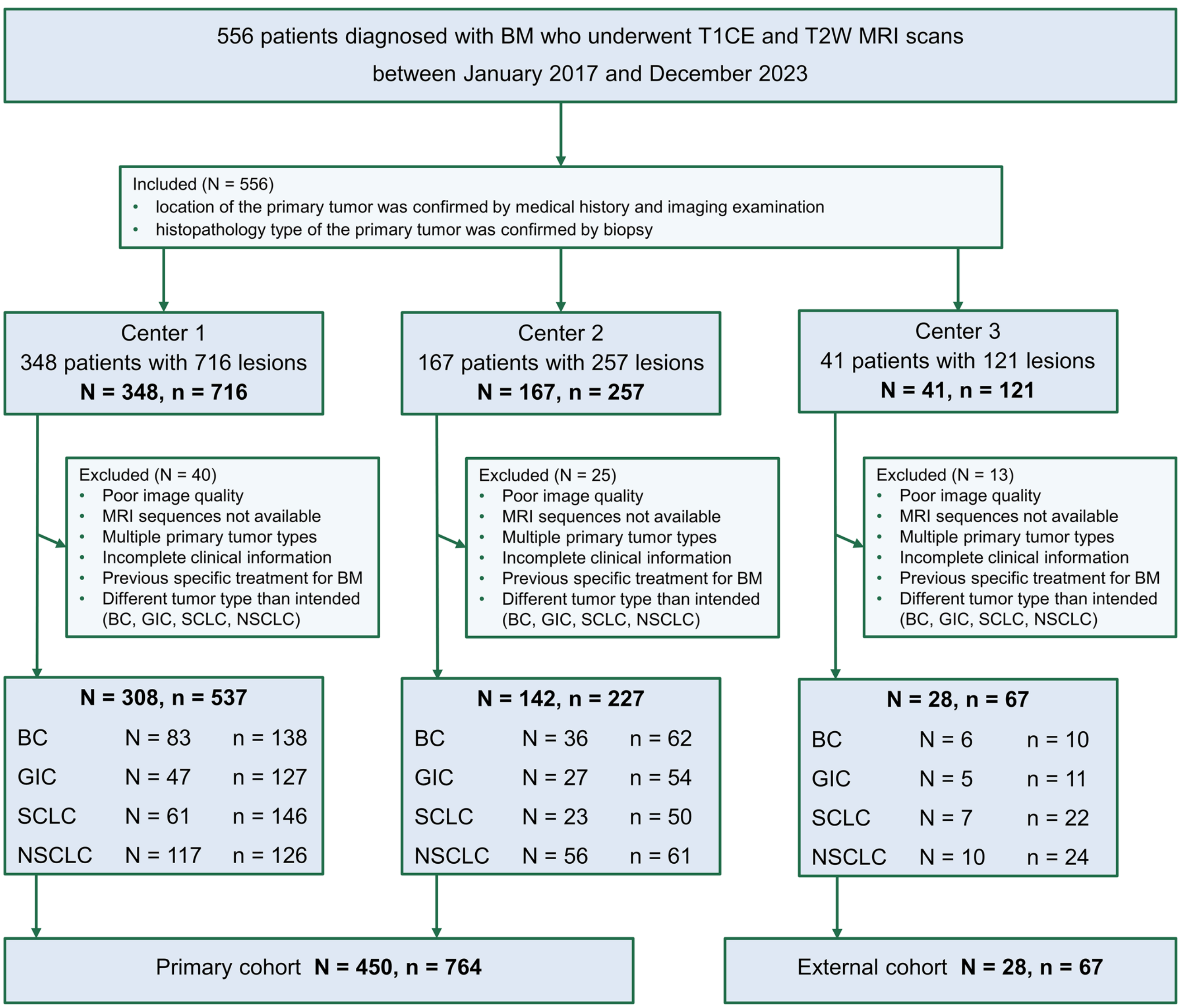

Fig. 1 Process for inclusion of eligible patients

into pixel units. Subsequently, the outer boundary of the manually delineated BM region is detected (using Python, Skimage.measure). This boundary is then adjusted by shifting 2 mm inward and 2 mm outward to generate a 4-mm-wide annular region (using Python, Skimage.draw). Volume of the peritumoral edema (VPE) was calculated using the ITK-Snap based on PEA. The script moves the tumor contour outward and inward by 2.0 mm separately and finally generates a ring-shaped region with a width of 4.0 mm.

Feature exaction and selection

In total, 3934 radiomics features were calculated from T1CE and T2W MRI (1967 for each single sequence) using the Pyradiomics package [21] in Python (version 3.8.3). The following different types of quantitative features are extracted from the brain MRI image: (i) Shape features include features that describe size of ROI or geometric properties of the volume of interest (VOI). (ii) First-order features respond the symmetry and homogeneity of the measured voxels, as well as changes in the local intensity distribution. (iii) Texture features include grayscale covariance matrix (GLCM) (the incidence of voxels with the same intensity value at a certain distance along a fixed direction), grayscale size zone matrix (GLSZM) (the distribution of groups of voxels with the same intensity), gray-level run length matrix (GLRLM), neighboring gray tone difference matrix (NGTDM), and gray-level dependence matrix (GLDM) features. The filtered features include square, local binary pattern (LBP) 2D/3D, Laplacian of Gaussian, gradient, wavelet, exponential, square root, and logarithm features. Details of the procedure for calculating radiomic features are provided in Appendix S1.

In order to screen for features that are with high correlation and low redundancy, all the extracted features were assessed by using Intraclass Correlation Coefficients (ICC) [22] analysis to remove the features with low confidence coefficients and retain the features that were with ICC > 0.85. First, to identify significant features ($p < 0.05$), Mann–Whitney U test was applied to the extracted features using the function wilcox.test in the R built-in package stats. Next, the least absolute shrinkage and selection operator (LASSO) algorithm was used to reduce the dimensionality of the features and performs tenfold cross-validation to retain the most important features [23].

Binary classification model construction

Radiomics signatures (RSs) were constructed using the selected most important features that are linearly combined with their respective weights. Finally, a binary logistic regression classifier was used to construct binary classification models for predicting metastatic types of BM based on the RSs. To compare the value of the BTI zone and the tumor region of BM, BTI-RSs and the BM-RSs were developed based on the selected features from BTI and BM, respectively. RSs based on the BTI zone were built to distinguish metastatic origins between lung and non-lung (RS-BTI-LC/NLC), SCLC and NSCLC (RS-BTI-SC/NSC), and BC and GIC (RS-BTI-BC/GIC). Similarly, RSs based on the tumor region of BM were built to distinguish metastatic origins between lung and non-lung (RS-BM-LC/NLC), SCLC and NSCLC (RS-BM-SC/NSC), and BC and GIC (RS-BM-BC/GIC). The VPE clinical indicator was used to build clinical models (VPE-LC/NLC, VPE-SC/NSC, VPE-BC/GIC). The combined radiomics models (Com-LC/NLC, Com-SC/NSC and Com-BC/GIC) were built by combining the developed RS-BTI-LC/NLC, RS-BTI-SC/NSC, and RS-BTI-BC/GIC with VPE-based clinical models.

Additionally, to explore the potential confounding effect of systemic treatment on BM imaging in patients, we conducted an additional stratified analysis. Patients were divided into Group 1 and Group 2 based on whether they had received systemic treatment before MRI acquisition. We then randomly selected 50 BM lesions from each group to assess the model performance.

Four-class classification model construction

A four-class model integrated four primary tumor types covering approximately 70%–90% of all BM incidences [3]. The comprehensive four-class classification model was constructed using the most valuable radiomic features obtained through feature extraction and selection, which utilized a multinomial logistic regression classifier to perform the four-class classification (SCLC vs. NSCLC vs. BC vs. GIC).

Statistical analysis

The statistical analysis was performed with R (version 4.2.2). Continuous variables were compared using the Mann–Whitney U test, while categorical variables were analyzed using the Chi-Square test. A two-sided $p < 0.05$ was considered statistically significant. The receiver operating characteristic (ROC) curves were plotted, and the area under the ROC curve (AUC) was calculated for evaluating the models. Delong's test was performed to evaluate differences among AUCs of different models. The corresponding specificity and sensitivity were determined by cutoff values based on the positive likelihood ratio. Figure 2 shows the design of this study.

Radiologist reading

An expert reader validation experiment was conducted with patients enrolled from the external test cohort. Four radiologists who have 2, 10, 15, and 20 years working

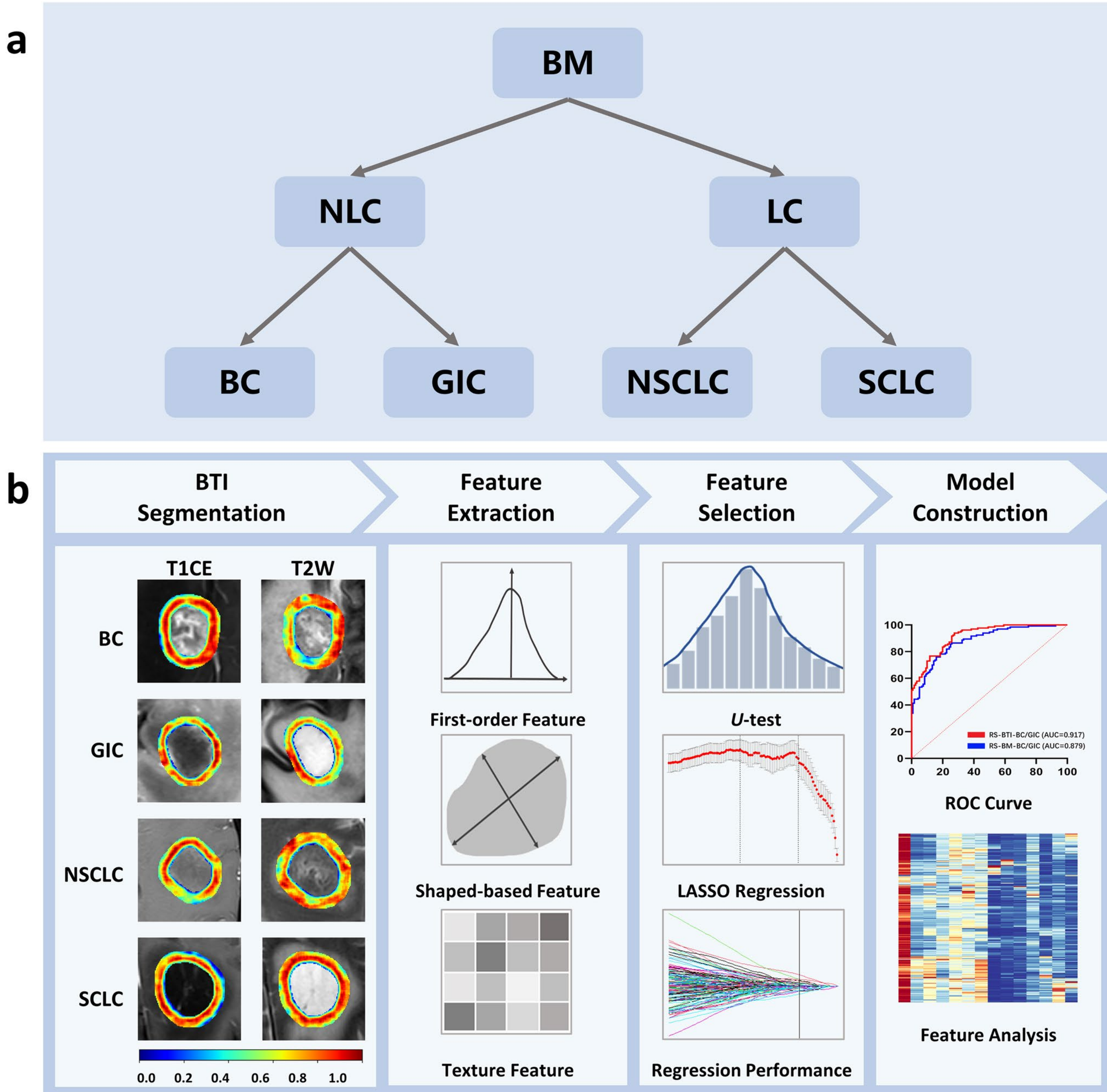

Fig. 2 Two-level stepwise architecture for the classification of the metastatic tumor types of BM (**a**). In the flowchart of the BTI-based radiomics analyses (**b**), the BTI segmentation section displays the entropy map of BTI, with blue indicating lower entropy (less informative) and red indicating higher entropy (more informative)

experiences were invited to predict the metastatic types of BMs based on following information: unprocessed T1CE and T2W MRI, patient's age and gender, and primary-specific MRI morphological features of the metastases. The MRI morphological features can be visually identified by radiologists, as listed in Table S5. The radiologists were blinded to the basic facts of the metastatic types of BMs and made their predictions individually. Specificity and sensitivity were calculated for evaluating the diagnostic capability of radiologists and the proposed radiomics models.

Results

Patients

A total of 478 patients were enrolled in this study, including 274 cases with lung-originated (91 of SCLC and 183 of NSCLC) and 204 cases with non-lung-originated (125 of BC and 79 of GIC) BMs. Table 1 shows the basic clinical characteristics of the patients, which showed no significant difference ($p > 0.05$) in gender and age on predicting the origin of BM.

Assessment of RSs derived from BTI and BM

LASSO selection results are shown in Figure S1, and formula of each developed RS is shown in Appendix S2. A total of 19, 14, and 16 radiomics features from the BTI zone were retained as the most important features for building RS-BTI-LC/NLC, RS-BTI-SC/NSC, and RS-BTI-BC/GIC, respectively. Tables S1, S2 and S3 show performance of the selected most important features. Table 2 lists the performance of the developed RSs for predicting the metastatic tumor types of BM. The RSs based on BTI region (RS-BTI-LC/NLC, RS-BTI-SC/NSC, and RS-BTI-BC/GIC) consistently performed superior than those based on BM regions (RS-BM-LC/NLC, RS-BM-SC/NSC, and RS-BM-BC/GIC) in terms of AUC and accuracy in both the primary validation and external test cohorts. This may indicate that the BTI region may hold more information correlated with metastatic types than the BM region. The results of Delong' tests in Table S4 showed a significant difference only between RS-BTI-LC/NLC and RS-BM-LC/NLC in the training, internal validation, and external test cohort. Figure 3 depicts ROC curves for each model in the primary validation and external test cohorts. However, the ROC curves of the other RSs (RS-BTI-BC/GIC vs. RS-BM-BC/GIC, RS-BTI-SC/NSC vs. RS-BM-SC/NSC) showed differences in AUC values, but these differences were not statistically significant. This may be due to the larger sample size in the LC/NLC models based on BTI and BM, which resulted in the AUC differences reaching statistical significance [24].

Assessment of combined models

The developed RS-BTI-LC/NLC, RS-BTI-SC/NSC, and RS-BTI-BC/GIC were combined with volume of peritumoral edema (VPE) to build combined radiomics models, named Com-LC/NLC, Com-SC/NSC, and Com-BC/GIC, respectively. As shown in Table 3, the VPE-LC/NLC, VPE-SC/NSC, and VPE-BC/GIC yielded AUCs ranging from 0.518 to 0.858 in the training, internal validation, and external

Table 1 Clinical characteristics of the patients in primary validation and external test cohorts

Characteristic	Cohort	Tumor type		*P*	Tumor type		*P*	Tumor type		*P*
		LC	NLC		SCLC	NSCLC		BC	GIC	
Patients	Primary	257	193		84	173		119	74	
	External	17	11		7	10		6	5	
Lesions	Primary	383	381		196	187		200	181	
	External	46	21		22	24		10	11	
Ages (Mean ± SD)	Primary	57.84 ± 9.68	61.24 ± 9.46	< 0.01*	59.44 ± 8.74	57.16 ± 8.39	0.06	54.49 ± 10.15	61.24 ± 9.46	< 0.01*
	External	63.00 ± 9.80	62.82 ± 12.74	0.98	60.86 ± 14.97	64.50 ± 3.89	0.88	62.00 ± 8.02	63.80 ± 17.98	0.83
Gender										
Male	Primary	153 (57.3%)	50 (25.9%)	0.51	54 (64.3%)	89 (51.4%)	0.05	0	50 (94.3%)	< 0.01*
	External	7 (41.2%)	2 (18.2%)	0.10	3 (42.9%)	4 (40.0%)	0.91	0	2 (40.0%)	0.03*
Female	Primary	114 (42.7%)	143 (74.1%)		30 (35.7%)	84 (48.6%)		119 (100.0%)	24 (5.7%)	
	External	10 (58.8%)	9 (81.8%)		4 (57.1%)	6 (60.0%)		6 (100.0%)	3 (60.0%)	

SD standard deviation; *P* values were calculated using Mann–Whitney U test for continuous variables (age) and Chi-square test for categorical variables (gender); *, $p < 0.05$

Table 2 Performance of models developed based on BM and BTI for identifying the metastatic tumor types of BM between LC and NLC, SC and NSC, and BC and GIC

RS	Training cohort				Internal validation cohort				External test cohort				
	AUC (95%CI)	ACC	SPE	SEN	AUC (95%CI)	ACC	SPE	SEN	AUC (95%CI)	ACC	BA	SPE	SEN
RS-BTI-LC/NLC	0.802 (0.765–0.840)	0.719	0.729	0.736	0.716 (0.653–0.779)	0.639	0.609	0.756	0.711 (0.576–0.846)	0.672	0.718	0.674	**0.762**
RS-BM-LC/NLC	0.724 (0.680–0.769)	0.652	0.659	0.659	0.645 (0.576–0.713)	0.576	0.641	0.617	0.646 (0.505–0.787)	0.672	0.694	0.674	0.714
RS-BTI-SC/NSC	**0.948 (0.923–0.973)**	**0.870**	0.838	**0.944**	**0.852 (0.788–0.916)**	0.752	0.864	0.698	**0.833 (0.709–0.958)**	**0.804**	**0.807**	0.864	0.750
RS-BM-SC/NSC	0.861 (0.815–0.906)	0.799	**0.869**	0.758	0.791(0.715–0.866)	0.682	0.667	**0.778**	0.746 (0.604–0.889)	0.652	0.716	0.682	0.750
RS-BTI-BC/GIC	0.917 (0.886–0.949)	0.822	0.725	0.940	0.839 (0.764–0.912)	**0.797**	0.852	0.776	0.791 (0.583–0.999)	0.667	0.805	**0.909**	0.700
RS-BM-BC/GIC	0.879 (0.838–0.919)	0.794	0.750	0.857	0.789 (0.680–0.845)	0.688	**0.902**	0.582	0.773 (0.565–0.980)	0.762	0.759	0.818	0.700

AUC area under the curve, *CI* confidence interval, *ACC* accuracy, *SPE* specificity, *SEN* sensitivity, *BA* balance accuracy, *RSs* radiomics signatures

The best results have been highlighted in bold

test cohorts. The generated Com-LC/NLC, Com-SC/NSC, and Com-BC/GIC showed an overall improvement in AUCs (0.717–0.949) compared with the BTI-based RSs without VPE in the training, internal validation, and external test cohorts. Table S4 shows that no significant difference was found in VPE versus Com-SC/NSC and VPE versus Com-BC/GIC in the primary training, internal validation, and external test cohorts. However, a significant difference was observed in VPE versus Com-LC/NLC only in the external test cohort, which may be related to the data distribution of the external test cohort. ROC curves of the Com-LC/NLC, Com-SC/NSC, and Com-BC/GIC are shown in Fig. 4.

Table S6 shows the results of the stratified analysis. The results indicated that three models' performances remain robust in both the treated Group 1 (AUC: 0.692–0.782; ACC: 0.580–0.645; SPE: 0.625–0.746; SEN: 0.667–0.710) and the untreated Group2 (AUC: 0.681–0.744; ACC: 0.687–0.700; SPE: 0.700–0.870; SEN: 0.556–0.698), with no significant difference observed between Group 1 and Group 2 ($p > 0.05$). These findings suggest that the models are robust to the effects of these treatments, and demonstrate that the models we developed are generalizable and can be applied to a broad patient population, including those who have undergone systemic treatments. Previous studies predicting the origin of BM have not considered the impact of systemic treatments on MRI imaging [11, 12], which enhanced the potential clinical utility of our model across diverse clinical scenarios.

Radiologist reading

Table 4 lists the performance differences between the radiologists reading and radiomics model. All four radiologists generated low diagnostic sensitivity and specificity for distinguishing metastatic origins of LC from NLC (SEN: 0.500–0.543; SPE: 0.564–0.646), SCLC from NSCLC (SEN: 0.524–0.600; SPE: 0.521–0.600), and BC from GIC (SEN: 0.597–0.667; SPE: 0.574–0.724). Notably, the radiologists generated the highest sensitivity (0.597–0.667) and specificity (0.574–0.724) for distinguishing metastatic origins between BC and GIC. Nevertheless, radiologists had difficulty distinguishing metastatic origins between LC and NLC, generating the lowest sensitivity (0.500–0.543). The established imaging-based radiomics models outperformed all radiologists in terms of sensitivity and specificity for distinguishing metastatic origins between LC and NLC (SEN: 0.762; SPE: 0.674), SCLC and NSCLC (SEN: 0.750; SPE: 0.864), and BC and GIC (SEN: 0.700; SPE: 0.909).

Assessment of four-class classification model

Table 5 lists prediction results of the comprehensive four-class classification model based on BM and BTI. By

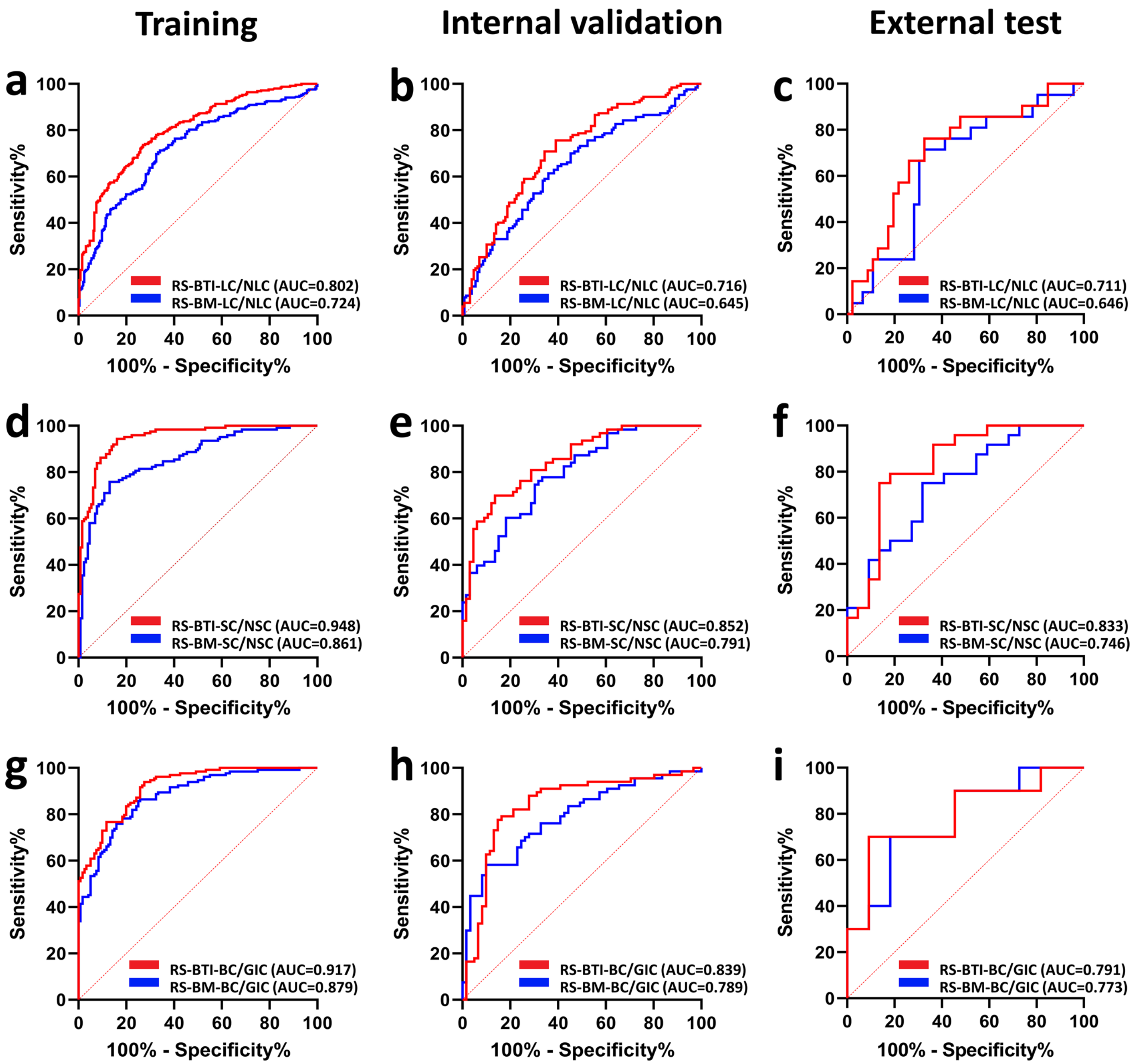

Fig. 3 ROC curves of the developed RS-LC/NLC, RS-SC/NSC and RS-GIC/BC in the training (**a**, **d**, **g**), internal validation (**b**, **e**, **h**), and external test (**c**, **f**, **i**) cohorts

comparing Tables 2 and 5, the results indicated that binary classification models outperformed the four-class classification model in terms of AUC values. The improved results show that binary classification models were more effective in correctly identifying classes and minimizing errors, making them more advantageous for the given cohort. This may be explained considering the simplicity and clarity of binary classification models which allows them to focus on distinguishing one class from another class (LC vs NLC, BC vs GIC, and SCLC vs NSCLC), which may contribute to their superior performance.

Discussion

Our study suggested that the proposed radiomics model may be used to predict the metastatic tumor type in patients with BM, including those whose primary tumor site remains unknown, which could potentially guide the prioritization of diagnostic evaluations [25]. Previous clinical studies showed that BM from different origins differs in tumor microenvironment [11, 26–28]. Brain MRI-based radiomics were proved to be helpful for noninvasively

Table 3 Performance of the Com-LC/NLC, Com-SC/NSC, and Com-BC/GIC for identifying the metastatic tumor types of BM

RS	Training cohort				Internal validation cohort				External test cohort			
	AUC (95%CI)	ACC	SPE	SEN	AUC (95%CI)	ACC	SPE	SEN	AUC (95%CI)	ACC	SPE	SEN
VPE-LC/NLC	0.518 (0.462–0.562)	0.505	0.541	0.504	0.586 (0.516–0.656)	0.502	0.773	0.929	0.561 (0.538–0.562)	0.731	0.826	0.619
Com-LC/NLC	0.803 (0.766–0.841)	0.715	0.725	0.740	0.717 (0.654–0.780)	0.639	0.610	0.756	0.744 (0.611–0.878)	0.731	0.652	0.810
VPE-SC/NSC	0.710 (0.647–0.773)	0.598	0.769	0.556	0.697 (0.607–0.787)	0.597	0.652	0.651	0.858 (0.747–0.969)	0.609	0.864	0.750
Com-SC/NSC	0.949 (0.924–0.974)	0.870	0.862	0.919	0.854 (0.791–0.917)	0.760	0.833	0.714	0.839 (0.714–0.937)	0.783	0.818	0.791
VPE-BC/GIC	0.561 (0.488–0.634)	0.518	0.700	0.910	0.560 (0.645–0.664)	0.523	0.590	0.821	0.672 (0.581–0.728)	0.619	0.545	0.600
Com-BC/GIC	0.918 (0.887–0.950)	0.822	0.725	0.940	0.840 (0.765–0.914)	0.797	0.836	0.791	0.800 (0.597–0.999)	0.666	0,909	0.700

AUC area under the curve, *CI* confidence interval, *ACC* accuracy, *SPE* specificity, *SEN* sensitivity, *RSs* radiomics signatures

identifying the origin of BM [11, 12], however, only focused on the tumor region of BM, and neglected the potential value of brain-to-tumor interface (BTI). Previous works have indicated that MRI features derived from BTI enable prediction of brain invasion in meningioma [15, 29]. Fan et al. [19] demonstrated that BTI-based features were associated with epidermal growth factor receptor (EGFR) mutation status and response to EGFR-tyrosine kinase inhibitors (TKI) in NSCLC-originated BM, which indicated that the BTI may hold information correlated with instinct microenvironment of BM. However, whether the BTI can reflect differentiated characteristics of BMs with different metastatic tumor types is not clear.

In this study, we evaluated two MRI sequences (T1CE and T2W) and compared RSs derived from BTI and BM, and found that BTI-based RSs always outperformed BM-RSs in terms of AUC and accuracy for distinguishing LC and NLC, NSCLC and SCLC, and BC and GIC. Compared with the whole tumor, the BTI region between tumor and brain parenchyma is an area where tumor cells interact with the intracranial immune system [14], showing structural and functional changes in the BM microvasculature and peritumoral edema [30]. Previous works [31, 32] indicated that metastatic cells in BM are infiltrative, expanding outward through brain capillaries and invading other tissues; thus, the area surrounding the whole tumor may contain a wealth of information. This may be an explanation to our results considering that the BTI region may contain more information than the whole tumor. Compared with radiologists' reading results, the proposed BTI-based RSs demonstrated improved performance in predicting metastatic types of BM, which may also suggest the potential of BTI-based radiomics.

The radiologists generated the highest predictive performance for distinguishing metastatic origins between BC and GIC, probably because gender-related factors, as BC cases occur almost exclusively in women. Additionally, the presence of specific morphological features in patients with GIC, such as hypointense masses on T2W MRI, may have further contributed to this distinction [25].

We identified 19 (RS-BTI-LC/NLC), 14 (RS-BTI-SC/NSC), and 16 (RS-BTI-BC/GIC) features from the BTI region that were closely related to the origin of BM. The majority of the features (35 of 49) belong to the category of textural features, which suggested that the heterogeneity within the MRI image of BTI contributes the predictive power for identifying the origin of BM. This was partially in line with Monika et al. [33] who showed that calculation of heterogeneity by texture analysis can reveal structural differences between lung- and BC-originated BMs.

The peritumoral edema (PE) of BM has been shown to represent unique microenvironment, known as vasogenic edema on MR. Although primary brain tumor and BM both can produce PE, there are differences in the molecular

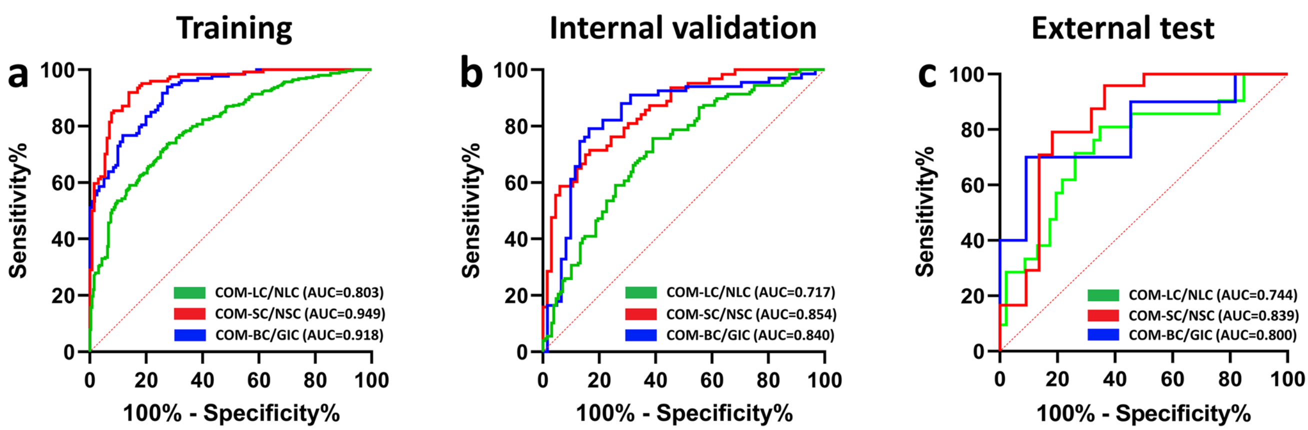

Fig. 4 ROC curves of the developed Com-LC/NLC, Com-SC/NSC and Com-BC/GIC in the training (**a**), internal validation (**b**) and external test (**c**) cohorts

Table 4 Comparison of reading and radiomics-based identification of metastatic types of BM

Tumor type	Reader 1		Reader 2		Reader 3		Reader 4		Radiomics model	
	SPE	SEN	SPE	SEN	SPE	SEN	SPE	SEN	SPE	SEN
LC/NLC	0.635	0.543	0.616	0.521	0.646	0.500	0.564	0.500	0.674	0.762
SC/NSC	0.600	0.554	0.574	0.535	0.574	0.600	0.521	0.524	0.864	0.750
BC/GIC	0.667	0.667	0.616	0.597	0.724	0.659	0.574	0.600	0.909	0.700

The radiologists with 2 years of experience for reader 1, 10 years of experience for reader 2, 15 years of experience for reader 3, and 20 years of experience for reader 4, respectively

Table 5 Performance of four-class classification models developed based on BM and BTI for identifying the metastatic tumor types

Tumor type	RS-BM								RS-BTI							
	Training cohort				Validation cohort				Training cohort				Validation cohort			
	AUC	ACC	SPE	SEN	AUC	ACC	SPE	SEN	AUC	ACC	SPE	SEN	AUC	ACC	SPE	SEN
BC	0.723	0.583	0.847	0.525	0.714	0.639	0.735	0.607	0.843	0.583	0.767	0.783	0.577	0.721	0.538	0.607
GIC	0.789	0.500	0.806	0.683	0.675	0.576	0.760	0.562	0.842	0.642	0.756	0.808	0.614	0.575	0.637	0.586
NSCLC	0.837	0.625	0.661	0.900	0.720	0.752	0.620	0.750	0.855	0.642	0.786	0.775	0.687	0.564	0.654	0.724
SCLC	0.780	0.592	0.753	0.683	0.563	0.682	0.756	0.582	0.837	0.592	0.692	0.833	0.674	0.552	0590	0.881

AUC area under the curve, *CI* confidence interval, *ACC* accuracy, *SPE* specificity, *SEN* sensitivity

mechanism during the development of PE in the two types of tumors [34]. Baris et al. [35] suggested that the VPE can be considered an independent factor in distinguishing between BM and primary brain tumor. Fan et al. [19] indicated the association of VPE and gene mutation status in NSCLC. Our study showed that the VPE alone can be predictive on the metastatic tumor types of BM with AUC ranging from 0.561 to 0.710. By integrating the VPE to BTI-based RS, the combined model significantly improved the predictive capability. Delong' test ($p<0.05$) showed that imaging-based features and VPE may hold complementary information regarding to the metastatic types of BM, which may widen the understanding in this field.

There are limitations in this study. First, the study failed to enroll patients with malignant melanoma (MM)-originated BM due to insufficient data. This is because the annual incidence of MM in Asian is less than 0.001%, which is much lower compared with the global average [36]. In the future, we aim to expand the melanoma patient dataset through multicenter collaborations to support further research. Second, although the study investigated the value of BTI for identifying the metastatic tumor types, it is not verified with histological sample. Third, this study only evaluated T1CE and T2W MRI; other MRI sequences such as non-contrast axial fluid-attenuated inversion recovery (FLAIR) should be assessed and may demonstrate further biological associations with the origin of BM. Fourth, although this study implied the potential correlation between the 4-mm BTI and the metastatic tumor type of BM, the study lacks determination of the optimal dilation distance to generate the ROI of BTI. Fifth, the developed model treated all lesions as independent, without adequately accounting for potential similarities among lesions from the same patient, such as the same primary tumor type and technical variables related to MRI scanning. Finally, manual segmentation of the whole tumor region and peritumoral edema is inefficient. Therefore, the application of algorithms such as deep learning-based segmentation technology should be considered in the future.

Conclusion

This study demonstrated the value of BTI and VPE for identifying the origin of BM. The developed MRI-based radiomics models may be helpful for radiologists to reveal the primary tumor and provide timely treatment.

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Declarations

Conflict of interest The authors of this manuscript declare no relationships with any companies, whose products or services may be related to the subject matter of the article.

Ethical approval This retrospective study was approved by the Institutional Review Board of the Cancer Hospital of China Medical University, Shengjing Hospital of China Medical University, and the People's Hospital of Liaoning Province, and the consents from patients were waived.

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